• Types of solids:-

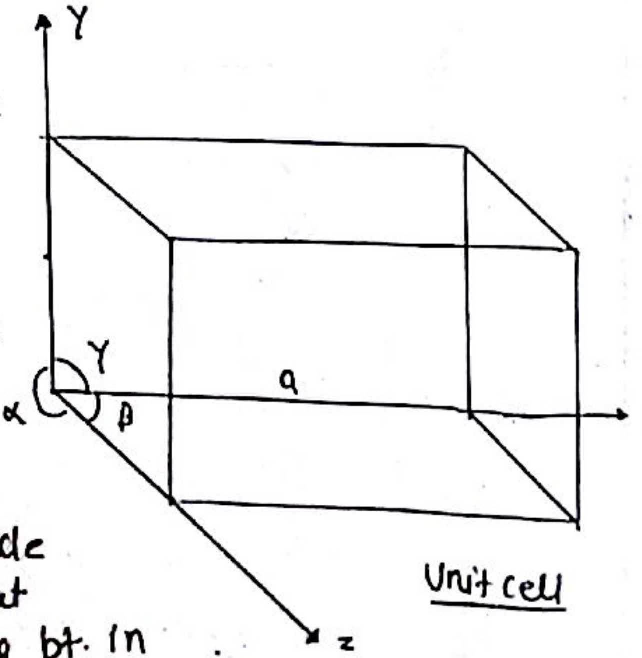
Solids → Crystalline solids, Amorphous Solids

• **Crystalline solids:** ⇒ crystalline solids are arranged in fixed geometric patterns or Lattices. They have oderly arranged units and are practically incompressible. Crystalline solids also show a definete melting point and so they pass rather sharply from solid to liquid state. The crystal are characterised by having a long range period in their st. and this gives the symmetry to the external st. for example:- Quartz, Silver, Cu, Au, Tourmaline.

→ **Amorphous Solids:** ⇒ A solid which lacks regularity in the arrangement of constituent particle are called Amorphous solids.
It means the pattern of its contituents atoms or molecules does not repeat periodically in the 3-D. Amorphous solids have short range order.
for example: glass, ruber, plastic, concreate.

• **Unit cell:** ⇒ The smallest portion of space lattice which can generate the complete crystal by repeating its own dimension in various directions is called a unit cell.

• **Space Lattice:** ⇒
A crystal such as rock salt, calcite, Quartz diamond etc is a solid compossed of geometrically regular arrangement of atoms, ions & molecules in the space. when it is broken into pieces, all pieces have the same shape with same angle b/w the corresponding faces. These angles are characteristics of the given crystal.



Unit cell

. Bravis assumed that a crystal is made up of 3-D array of points such that each point is surranded by neighbouring pt. in an identical ways. such an array of pt in 3-D is called Space Lattice or Bravis Lattice.
hence space Lattice is regular periodic arrangement of points extended repeatedly in space.

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Types of crystal :⇒ All crystals are classified into seven crystal system on the basis of the shape of the unit cells. Bravis in 1948 explained that there are fourteen different types of crystal lattice under the seven crystal system. These seven basic system are distinguish from one another by their axial ratio a, b, c & the angles b/w them. These angles are α, β and γ

o	Name of crystal	Relation in side	Angles	No. of possibles Lattice	Example
	cubic	$a=b=c$	$\alpha=\beta=\gamma=90^\circ$	3 (P, F, I)	NaCl
	Tetragonal	$a=b\neq c$	$\alpha=\beta=\gamma=90^\circ$	2 (P, I)	$NiSO_4$
	Orhorombic	$a\neq b\neq c$	$\alpha=\beta=\gamma=90^\circ$	4 (P, C, F, I)	KNO_3
	Monoclinic	$a\neq b\neq c$	$\alpha=\beta=90^\circ,\neq\gamma$	2 (P, C)	$FeSO_4$
	Triclinic	$a\neq b\neq c$	$\alpha\neq\beta\neq\gamma\neq90^\circ$	1 (P)	$CuSO_4$
	Trigonal	$a=b=c$	$\alpha=\beta=\gamma\neq90^\circ$	1 (R)	$CaSO_4$
	Hexagonal	$a=b\neq c$	$\alpha=\beta=90^\circ,\gamma=120$	1 (P)	Quartz

• P → Primitive, R – Rhomohedral C = Base centered, I = Body centered F = Face centered.

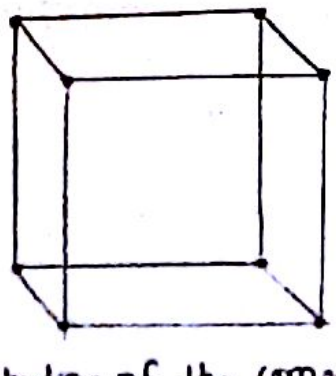
cubic :⇒ The cubic crystal have all sides are same all angles are 90°. It is divided into three subtypes

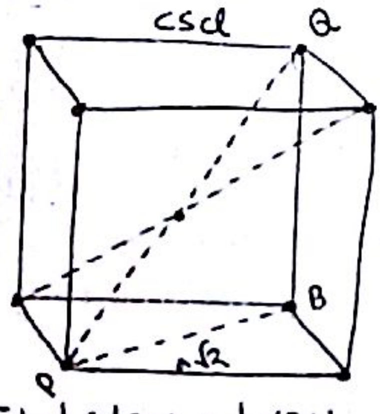
SCC → simple cubic crystal : (ii) BCC → Base centered cubic (iii) FCC

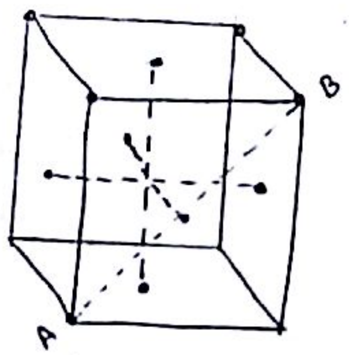
.. NH_4Cl, Pure Na, Li, Cr

CsCl

NaCl







contribution of the corner atom to the unit cell = $\frac{1}{8}$

Total no. of corner atoms = 8

Total contribution = $\frac{1}{8}\times 8 = 1$

In simple c.c lattice there is only one atom or molecule or ion per unit cell.

→ Total atom = $\frac{1}{8}\times 8+1$

$= \underline{\underline{2}}$

→ Total atom = $\frac{1}{8}\times 8$

$+\frac{1}{2}\times 6$

$= 1+3$

$= \underline{\underline{4}}$

⇒ Co-ordination no = Total atom present in nearest neighbouring atom

⇒ SCC

→ C.N = 6

$(\pm a\hat{i}, \pm a\hat{j}, \pm a\hat{k})$

$[\pm a,0,0]\ [0,\pm a,0]\ [0,0,\pm a]$

⇒ Distance b/w nearest neighbouring atom(r) =

$$2r = a$$

⇒ $\boxed{r = \frac{a}{2}}$

⇒ BCC

→ C.N = 8

$\pm\frac{a}{2}\hat{i}, \pm\frac{a}{2}\hat{j}, \pm\frac{a}{2}\hat{k}$

$\left(\frac{a}{2},\frac{a}{2},\frac{a}{2}\right)\left(-\frac{a}{2},\frac{a}{2},\frac{a}{2}\right)$

$\left(a/2,-a/2,a/2\right)\left(a/2,a/2,-a/2\right)$

$\left(-a/2,-a/2,a/2\right)\left(a/2,-\frac{a}{2},-\frac{a}{2}\right)$

$\left(-\frac{a}{2},-\frac{a}{2},-\frac{a}{2}\right)\left(-\frac{a}{2},\frac{a}{2},-\frac{a}{2}\right)$

⇒ $PQ = \sqrt{(a\sqrt{2})^2 + a^2}$

$$PQ^2 = 3a^2$$

$$(r+2r+r)^2 = 3a^2$$

$$(4r)^2 = 3a^2$$

$$4r = \sqrt{3}a$$

⇒ $\boxed{r = \frac{\sqrt{3}a}{4}}$

⇒ FCC

→ C.N = 12

$\left(\pm\frac{a}{2}\hat{i} \pm \frac{a}{2}\hat{j},\ \pm\frac{a}{2}\hat{j} \pm \frac{a}{2}\hat{k},\ \pm\frac{a}{2}\hat{k} \pm \frac{a}{2}\hat{i}\right)$

⇒ $AB = a\sqrt{2}$

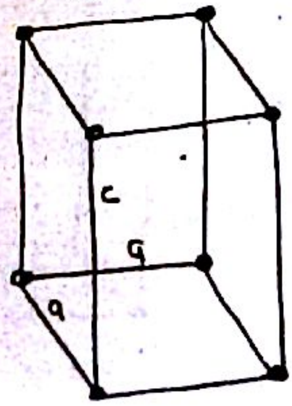
$$r+2r+r = a\sqrt{2}$$

$$4r = a\sqrt{2}$$

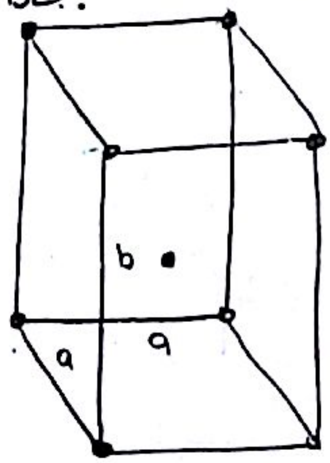
⇒ $\boxed{r = \frac{a}{2\sqrt{2}}}$

(ii) **Tetragonal crystals:** ⇒ These crystal have 2 sides are equal and all angles are equal. This crystal divided into 2 subtypes

(i) Perimitive



(ii) BC.



(iii) **Orthorhombic crystals:** ⇒
This crystals have all the side of different length but all the angles are 90°. This crystals are divided into 4 subtypes.

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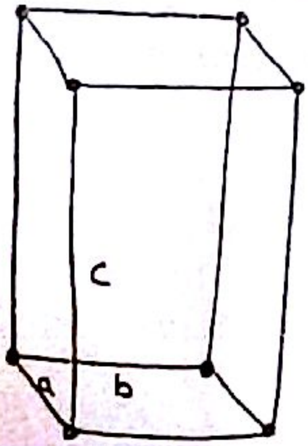
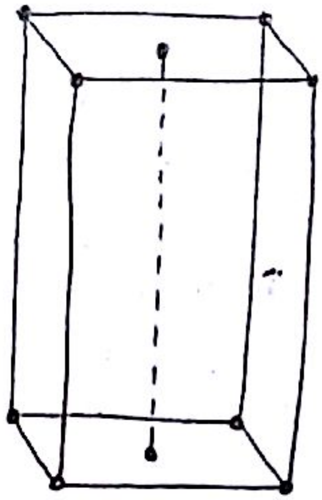
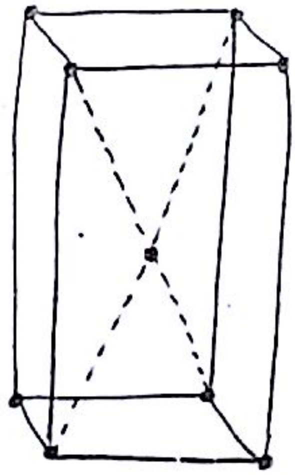
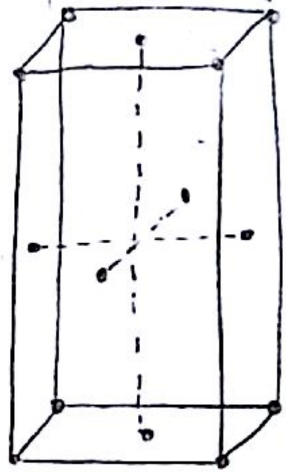
(i) Primitive (ii) Base centered (iii) Body centered (iv) Face centered



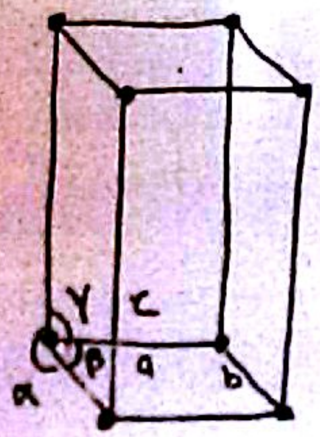




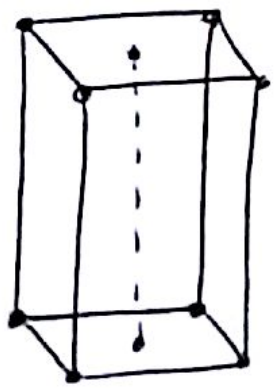


④ Monoclinic :→
Monoclinic crystal have all side are different in length but 2 . . angles are 90° and one angles are does not equal to 90°
These have two subtypes. ex- Sodium Borate

(i) Primitive



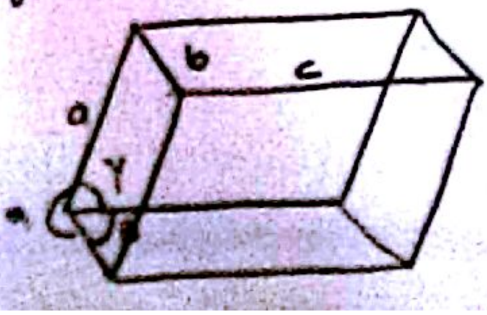
(ii) Base centered



$\alpha = \gamma = 90°$ $\beta \neq 90°$

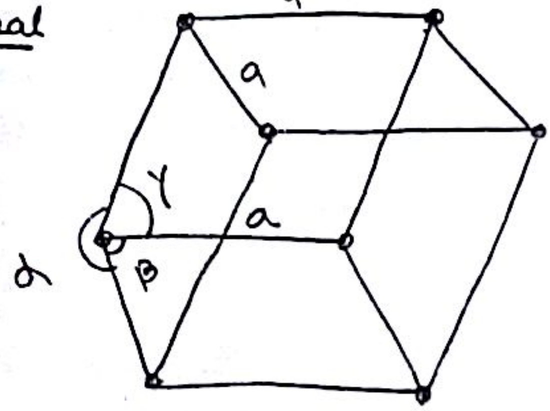
⑤ Triclinic :→ None of the axes are ⊥r to the other axis $\alpha \neq \beta \neq \gamma$ and also all sides are of different length.
These divid into subtypes. ex. $K_2Cr_2O_7$

① Primitive

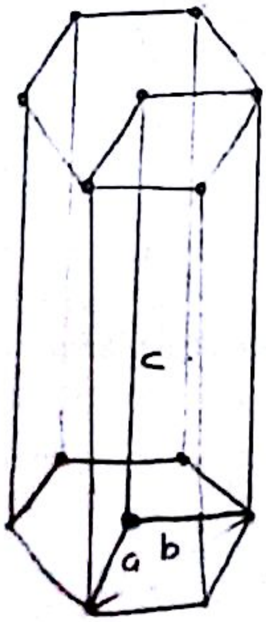


(6) **Trigonal** :⇒ All sides are equal in this type of crystal But two angle are equal to 90° And another angle does not equal to the 90° These crystal divided into 1 subtypes ex- Arsenic, Bismeth. Sb, $CaCO_3$

(i) Rhombohedral



(7) **Hexagonal crystals** :⇒ In this crystal two sides are equal but another side are of different length from two sides. Two angles are of 90 and one angle is 120° There dived into one subtypes.

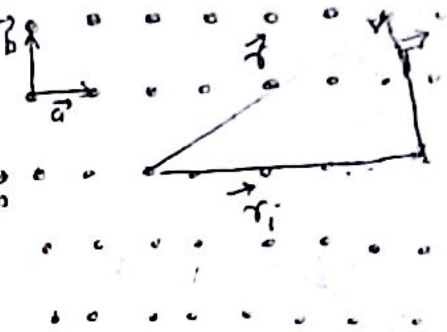


Using stewings approx

$\log_e x! = x \log x - x$

⇒ Translational Vectors: ⇒ Mathematically a lattice is define by 3 fundament-al vectors a, b and c vector such that the atomic arrangement looks exactly identical when viewed from any points $\vec{r}$ and $\vec{r_1}$

$$\vec{r_1} = \vec{r} + \vec{T}$$
$$= \vec{r} + n_1\vec{a} + n_2\vec{b} + n_3\vec{c}$$

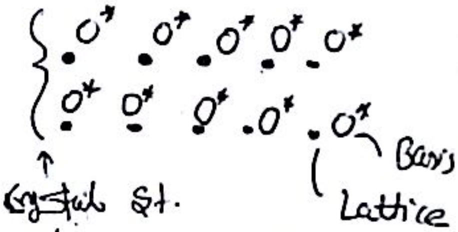


n_1, n_2 & n_3 are integer

$$\vec{T} = n_1\vec{a} + n_2\vec{b} + n_3\vec{c} \text{ in 3D} \Rightarrow \text{in 2D } \vec{T} = n_1\vec{a} + n_2\vec{b}$$

where a, b & c are primitive along the x, y & z axis respectively.

⇒ Crystal structure = Lattice + Basis
↓ Real ↓ imaginary (space)



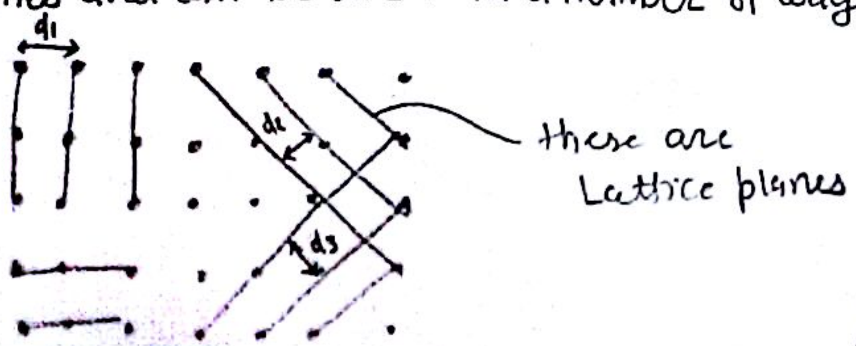
→ In the simplest crystal such as Cu, Ag Na, etc there is single atom or ion associated with each Lattice pt. more often we say that however there is a gr. of several atom associated with each Lattice pt. this gr. is called basis. Each basis is identical in composition, arrangement, orientation with any other basis

$+ \ O^{*} \rightarrow$ crystal st.

Lattice + basis

→ space Lattice is imaginary while (crystal st. (Lattic + basis)) is real.

→ **Lattice planes**: ⇒ A crystal lattice is made of a large no. of parallel equidistant planes is known as lattice planes and can be chosen in a number of ways as shown in fig.



→ A given space lattice may have an aggregate set of ~~1~~ equally spaced plane passing through the Lattice pt. this plane are called Lattice plane and the ⊥ distance b/w the adjacent plane is interplanar spacing.
As shown in fig. there 3 set of Lattice plane d_1, d_2 & d_3 are ⊥ distance b/w two plane respectively.

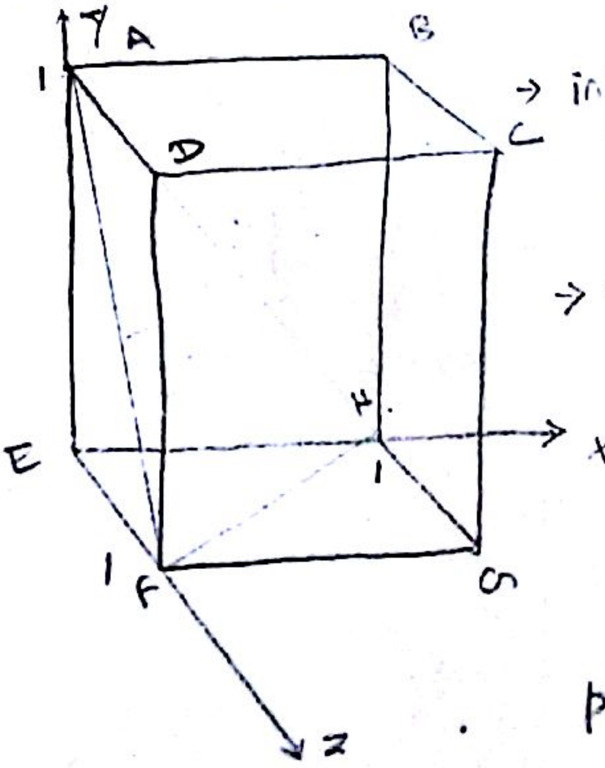
→ Each set of plane accumulates all the Lattice point. out of these. Only those which have high density of Lattice pt are significant and show diffraction of x-ray (Bragg's diffraction) these are called Bragg's plane or cleavage plane. when a crystal is struck. it is easily break across these plane.

* Explain Miller Indices ?

Imp ⇒ **Miller INDICES** :⇒ (Orientation of planes):

The integer which determine the orientation of a crystal plane in relation to three crystallographic axes are called Miller indices.
Miller indices are also called Crystal indices.

→ Orientation of these plane in the crystal are determined by 3 smallest whole no. which have same ratio's with one another as the reciprocal of the intercepts of the plane of the 3 crystal axis.



→ intercept on x → 1
" " y → 1
" " z → 1

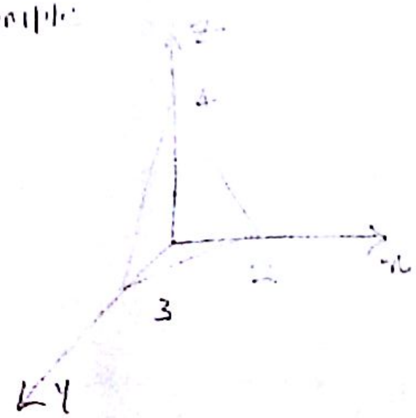
→ reciprocal of these intercept

$\frac{1}{1}, \frac{1}{1}, \frac{1}{1}$

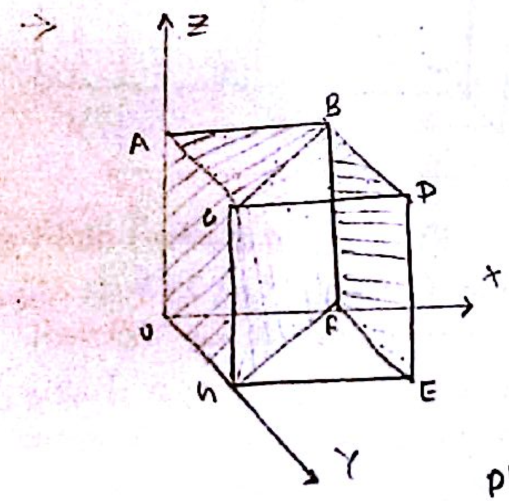
$(h, k, l) \equiv (1, 1, 1)$

plane $\underline{AHF}$

For example



$h,k,l \equiv (\ldots, 4, 3)$



plane → FBDE

$x \to 1$ $\quad \frac{1}{1}, \frac{1}{\infty}, \frac{1}{\infty}$

$y = \infty$ $\quad h,k,l \equiv (1,0,0)$

$z = \infty$

ACOG plane →

$x \to 0$ $\quad \frac{1}{0}, \frac{1}{1}, \frac{1}{1}$

$y \to 1$ $\quad (\infty\ 1, 1)$

$z = 1$

plane ABCD

$x \to \infty$ $\quad \frac{1}{\infty}, \frac{1}{\infty}, \frac{1}{1}$

$y \to \infty$ $\quad (0,0,1)$

$z = 1$

plane CBGF

$x \to 1$ $\quad \frac{1}{1}, \frac{1}{1}, \frac{1}{\infty}$

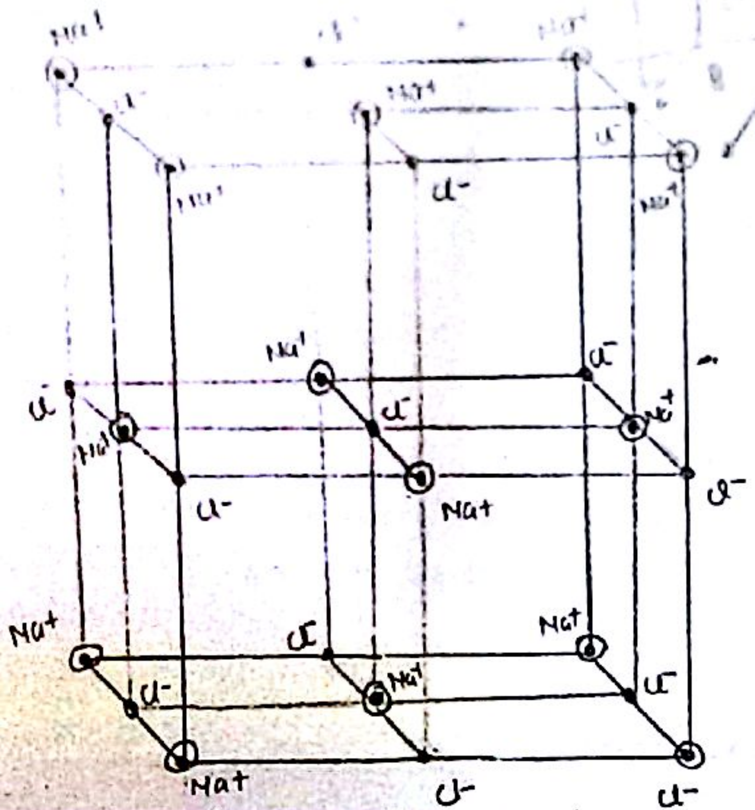
$y = 1$

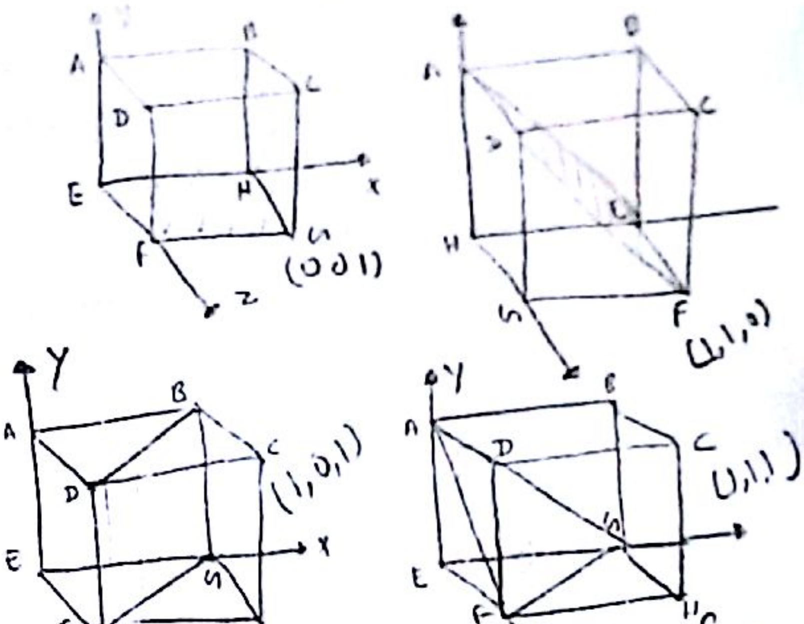
$z = \infty$

$h,k,l \equiv (1,1,0)$

Simple Crystal Structures:⇒

St of NaCl:⇒





⇒ Main features of NaCl St.

1)⇒ In NaCl St Na^+ ion are situated at corner as well as center of the faces. therefore we can say that Na^+ ion is FCC Lattice.

2)⇒ Each cell has 8 corners and 8 cell meet at the each of the corner hence $\frac{1}{8}$ atom belong to each cell also cation at the center of face is shared by two faces by two cells therefore $\frac{1}{2}$ ion belong of each cell.

→ we have 8 corner and 6 Faces therefore NaCl Lattice has four ion of the same kind and 4 ion of other kind each Na^+ ion has 6 Cl^- ion

③⇒ The volume of each of unit cell is a^3 and the distance b/w 2 nearest neighbour $= a/\sqrt{2}$

(iv) Each of Na^+ is attracted by 6 nearest $Cl^{\ominus}$ neighbours

(v) The ex. of family KCl, LiH, PbS.

⇒ Determination of Lattice constant a from the molecular wt.

⇒ Relation b/w Density of crystalline material and Lattice constant.

if a = side of unit cell

vol. of unit cell $= a^3$

ρ = density of crystalline material

$$\text{density} = \frac{\text{mass}}{\text{vol.}}$$

$$\rho = \frac{\text{mass per unit cell}}{a^3}$$

$$\text{mass per unit cell} = a^3 \rho \quad \text{——} ①$$

M = molecular wt. of crystalline material

N = Avagadro's no.

$$\text{mass of each molecule} = \frac{M}{N} \quad \text{——} ②$$

If n (no. of molecule or atoms) is the no. per unit cell

$$\therefore \text{Total mass per unit cell} = n\frac{M}{N} \quad \text{——} ③$$

Comparing ① & ③

$$a^3 \rho = n\frac{M}{N}$$

$$\Rightarrow a^3 = \frac{nM}{N\rho}$$

$$\boxed{a = \left[\frac{nM}{N\rho}\right]^{1/3}}$$

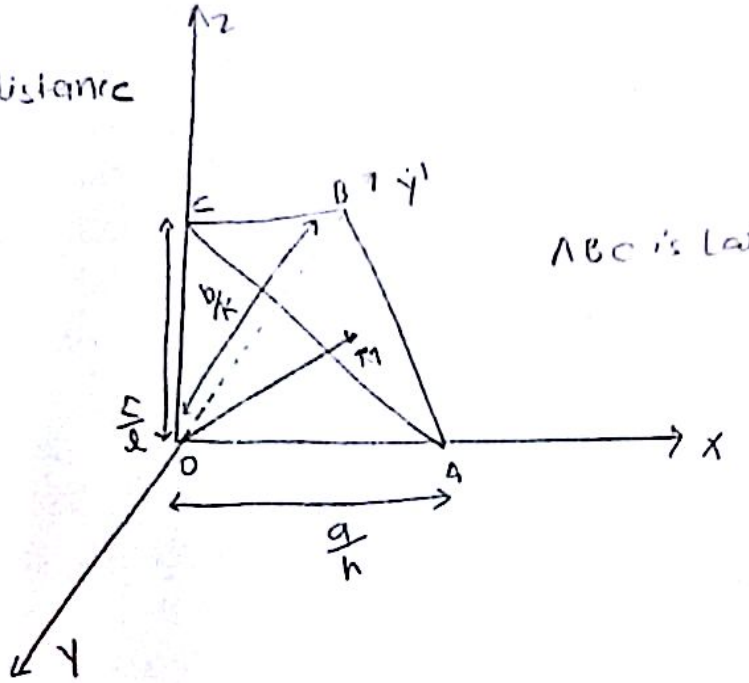
⇛ Derivation of interplanar spacing: ⇒

Interplanar spacing is the distance b/w two parallel lattice planes
Interplanar spacing b/w planes of cubic lattice

ON : Normal distance

= d

ABC is lattice plane



Let the normal makes an angle α, β, γ with 3 ⊥r axis and miller indices of the plane is , h, K, l

$$\cos\alpha = \frac{ON}{OA} = \frac{d}{a/h} = \frac{dh}{a}$$

$$\cos\beta = \frac{ON}{OB} = \frac{d}{b/k} = \frac{dk}{b}$$

$$\cos\gamma = \frac{ON}{OC} = \frac{d}{c/l} = \frac{dl}{c}$$

for direction cosines

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

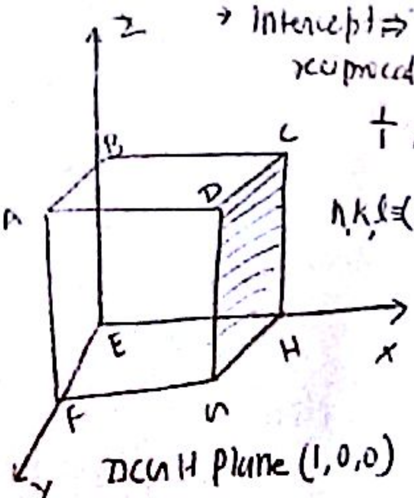
$$\frac{d^2h^2}{a^2} + \frac{d^2k^2}{b^2} + \frac{d^2l^2}{c^2} = 1$$

$$d^2\left[\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}\right] = 1$$

$$d = \frac{1}{\sqrt{h^2/a^2 + k^2/b^2 + l^2/c^2}}$$

For cubic simple Lattice $a=b=c$ $\therefore$ $\boxed{d = \frac{a}{\sqrt{h^2+k^2+l^2}}}$

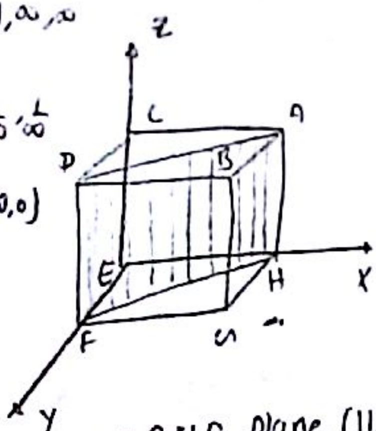
⇒ SCC



→ Intercept ⇒ $1,\infty,\infty$
reciprocal
$\frac{1}{1}, \frac{1}{\infty}, \frac{1}{\infty}$
$h,k,l = (1,0,0)$

DCGH Plane (1,0,0)

$d_{(1,0,0)} = \frac{a}{\sqrt{1+0+0}} = a$



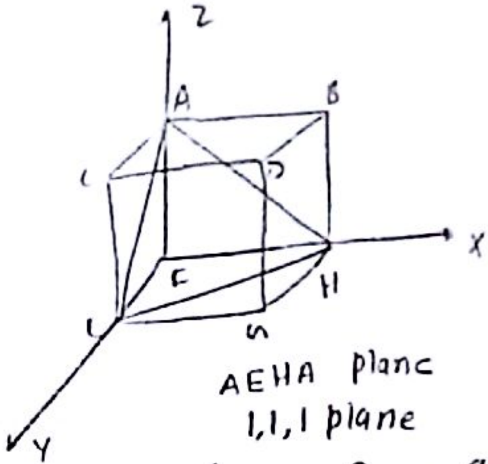
DAHF plane (1,1,0)

Intercepts $1,1,\infty$

$\frac{1}{1}, \frac{1}{1}, \frac{1}{\infty}$

$h,k,l = (1,1,0)$

$d_{(1,1,0)} = \frac{a}{\sqrt{1+1+0}} = \frac{a}{\sqrt{2}}$



AEHA plane

1,1,1 plane

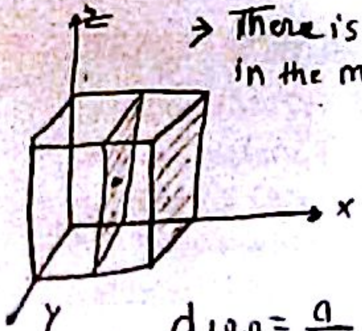
$d_{1,1,1} = \frac{a}{\sqrt{1+1+1}} = \frac{a}{\sqrt{3}}$

⇒ $d_{1,0,0} : d_{1,1,0} : d_{1,1,1} :: \frac{a}{1} : \frac{a}{\sqrt{2}} : \frac{a}{\sqrt{3}} :: 1 : \frac{1}{\sqrt{2}} : \frac{1}{\sqrt{3}} :: \boxed{1 : 0.71 : 0.58}$

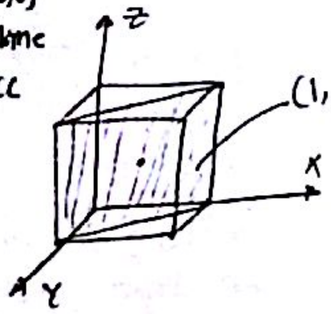
⇒ BCC –

(1,0,0)

→ There is additional plane in the midway of BCC

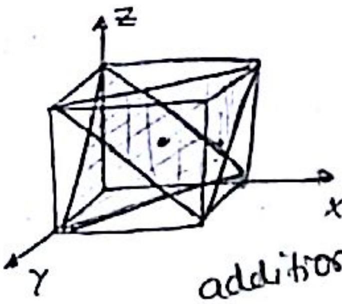


$d_{1,0,0} = \frac{a}{2}$



(1,1,0) plane

$d_{1,1,0} = \frac{a}{\sqrt{2}}$

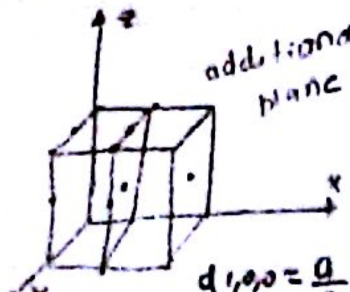


additional plane

$d_{1,1,1} = \frac{a}{2\sqrt{3}}$

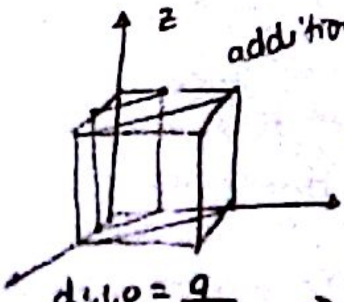
⇒ $d_{1,0,0} : d_{1,1,0} : d_{1,1,1} :: \frac{a}{2} : \frac{a}{\sqrt{2}} : \frac{a}{2\sqrt{3}} :: \boxed{1 : 1.414 : 0.58}$

⇒ FCC ⇒



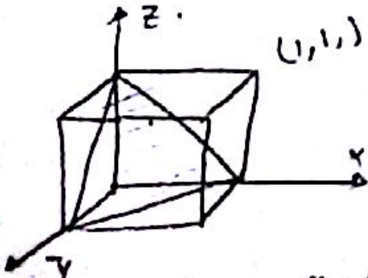
additional plane

$d_{1,0,0} = \frac{a}{2}$



additional

$d_{1,1,0} = \frac{a}{2\sqrt{2}}$



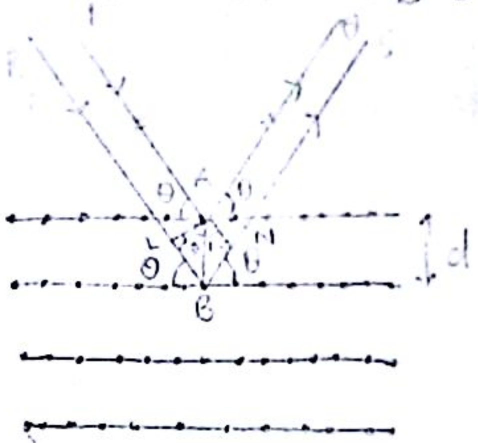
(1,1,1) $d_{1,1,1} = \frac{a}{\sqrt{3}}$

⇒ $d_{1,0,0} : d_{1,1,0} : d_{1,1,1} :: \frac{a}{2} : \frac{a}{2\sqrt{2}} : \frac{a}{\sqrt{3}} :: \boxed{1 : 0.71 : 1.15}$

⇒ Bragg's X- Ray diffractions: ⇒ When a narrow beam of x-rays of wavelength λ is incident on the parallel planes at angle θ. The x-rays scattered by the atoms of the crystal. Parallel Lattice planes of the crystal are separated by a distance d. The incident x-rays get diffracted at angle θ at point A and B of the crystal planes.

$AB = d$

Path difference b/w two defracted rays



Path difference = BL+BN

$= d\sin\theta + d\sin\theta$

$= 2d\sin\theta$

for maxima

$\boxed{2d\sin\theta = n\lambda}$ where $n = 1, 2, 3, 4$ - - -

d = Intaplaner spacing

θ = Glancing angle or Grazing angle

→ For 1st order

$n=1$

$2d\sin\theta = \lambda$

$\boxed{\sin\theta = \frac{\lambda}{2d}}$ //

This relation is known as Bragg's law of diffraction. It is useful in calculating the distance d between crystal Lattice planes once we know the wavelength λ of the X-rays. and measure the angles of diffraction θ

Bragg's condition :⇒

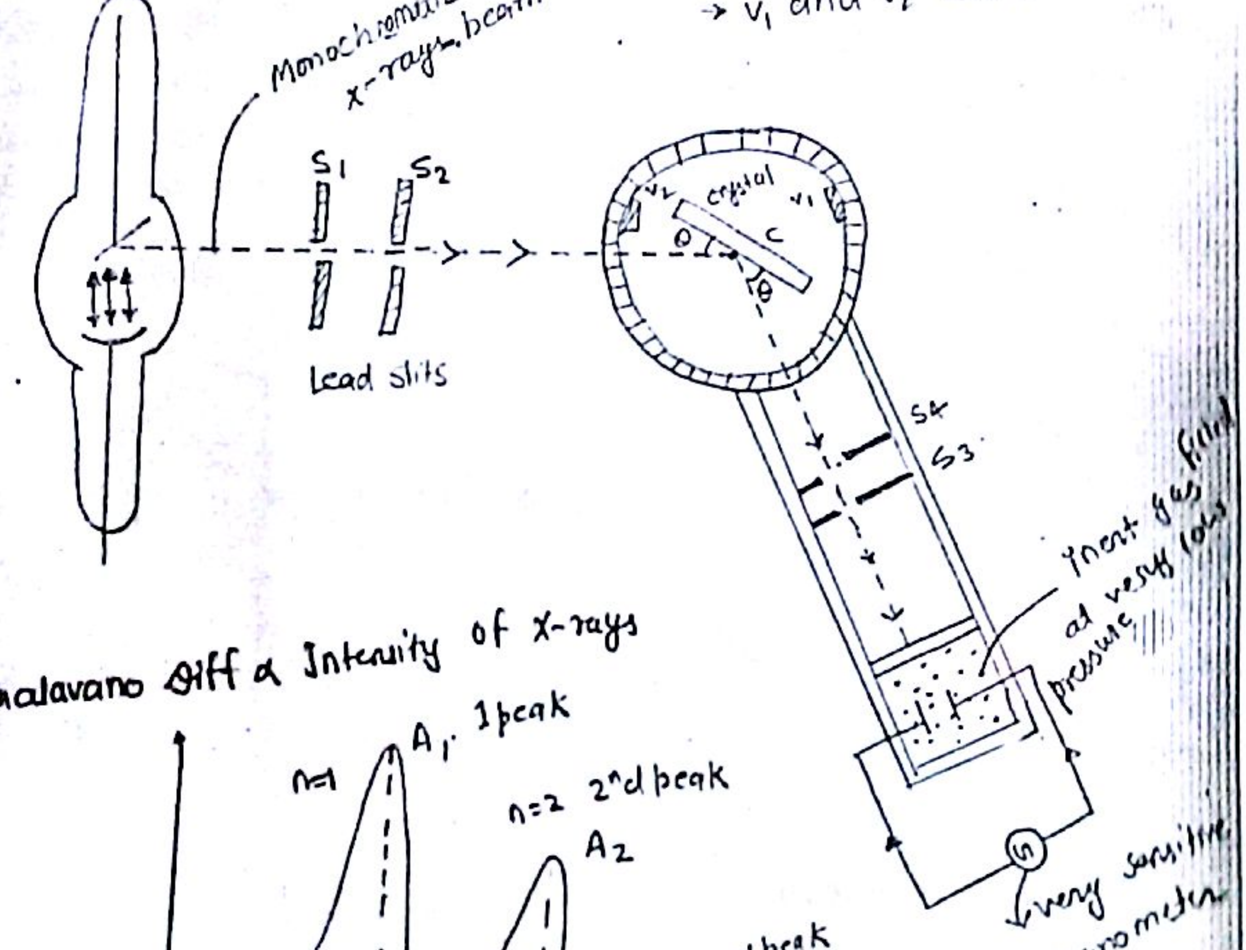
⇒ (i) That every x-ray will not get diffracted by the atom of a crystal. only those will be diffracted whose wavelength λ and the ange θ exactly match this condition.

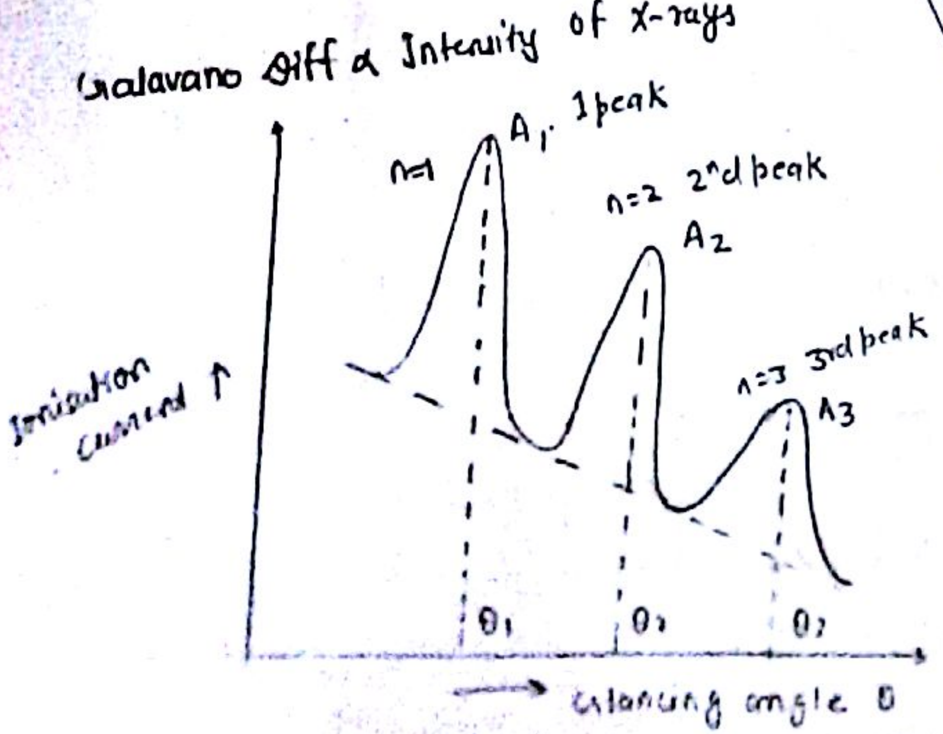
⇒ Bragg's X-ray spectrometer ⇒ (Laboratory experiment) :⇒

Bragg's x-ray spectrometer is used to study the glancing angle θ and intensities of diffracted x-rays for a given crystal.

A monochromatic x-ray beam from the source is passed through two slit S_1 and S_2 which olimate is into a fine narrow beam. This x-ray beam is allow to fall on crystal simple C mouted at the centre of a turn table. A movable arm is also attached to the turn table for detecting the deflected x-rays beam from the Crystal C. This turn table is capable of rotation about a vertical axis and the angle of rotation can be measured on the circular scale. The rates of rotation of the turn table and the detector (ionisation chamber) arm are such that the ionisation chamber always receives the diffracted beam. When the turn table rotates through an angle θ the ionisation chamber arm automatically rotates through an angle 2θ with the direction of incident ray. This way the measurement of different diffracted x-ray beams' intensities and angles are recorded. Then using Bragg's condition we get the interplanar spacing and the st. of the crystal.

→ V_1 and V_2 are verniers





for 1st Order $\lambda_1 \rightarrow 2d \sin\theta_1 = \lambda$

$n = 2 \quad \lambda_2 \rightarrow 2d \sin\theta_2 = 2\lambda$

$n = 3 \quad \lambda_3 \rightarrow 2d \sin\theta_3 = 3\lambda$

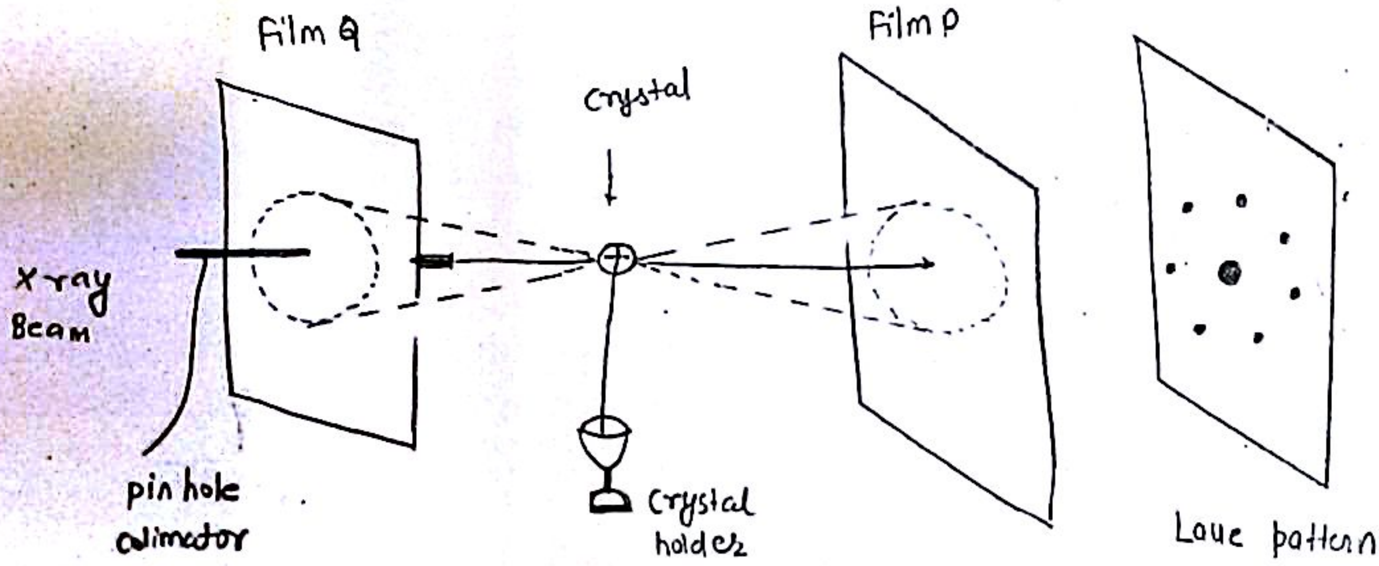
$$\sin\theta_1 : \sin\theta_2 : \sin\theta_3 = 1 : 2 : 3$$

⇒ Resuts : ⇒ (i) As the Order of spectrum inc. Intensity dec.

(ii) The ionisation current does not falls to zero for any range of glancing angle θ. It is max^m for certain value this shows the existance of continous spectrum on which the characteristics line spectrum is super possed.

⇒ Laue Method : ⇒

The Laue Method is useful for the determination of crystal structure. In this method a single crystal is held stationary in the path of an incident x-ray beam as shown in fig.



⇒ When a continous x-ray through pin hole is allowed to fall on the crystal then this beam is diffracted by the crystal the transmitted-diffracted and the reflected-diffracted beams are received by the films P and Q respectively as shown in fig.

The transmitted diffracted beams from a series of spots which is the characteristics of the crystal st. and is called Laue pattern.

Each dot on the Laue pattern correspond to the interference maxima for a set of crystal planes satisfying the Bragg's condition [illegible] for a particular wavelength selected from the beam of incident lights. By studying the position and intensities of these [illegible] the crystal str. can be determined.

⇒ crystal Defect: ⇒

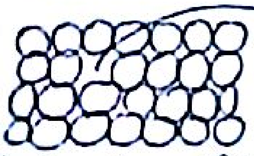
→ Ideal Crystal: ⇒ Any idealy perfect crystal is one in which atom and crystal are arranged in perfectly regular manner without any deviation. however it is impossible an ideal crystal without any defect in nature.

⇒ Defect: ⇒ Any deviation from the perfect periodically is called defect in the crystal.

⇒ Real crystal has some defects out of the following Defects:

(i) Point defect
(ii) Line Defect
(iii) surface Defect

→ (i) Point Defect: ⇒ The deviation from periodically regular arrangement the crystal is localised only around a few atom then this type of defect in crystal is called point defects.
This produce a very localised interaption in the regularity of the Lattice. The point defect produce strain in a very small region of a crystal around the defect but does not affect the regularity of distinct region. The pt. defect in a crystal may be in the form of of impurity or in the form of vacancy.

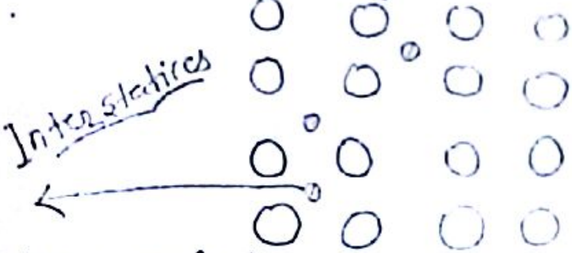


⇒ Vacancy ⇒ A vacany is the simplest pt. defect in the crystal this defect arises due to missing atom and vacant atomic site. The vacancy arises when the atom leave its site. such defect are either due to the imperfect packing or during original crystalisation or form thermal vibration of the atoms at high temp^r.

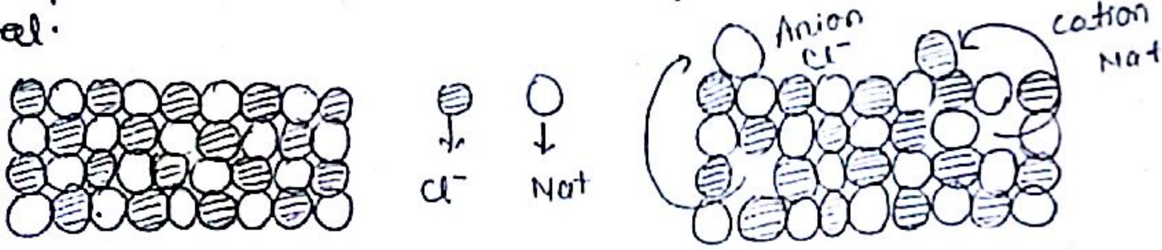
→ when the crystal is heated, the probability of the atom which jumps out of their position will res. for most of the case the thermal energy is ater of 1ev per vacancy. This vacancy may be single or double

⇒ Interstatial ⇒ when atomic packing factor is low in a crystal st. an extra atom may be introduce in the crystal st. This extra atom can take the interstatial space b/w the regularly arranged atom when it is smaller than the parent atom. It will produce atomic destortion hence the defect in which the extra atom enters in a interstatial space b/w the regular arranged atom of losely packed crystal (s) It is known as Interstatial.

- It is clear that vacancy and interstatial are opposite to the each other.



⇒ **Schottkey Defects**: ⇒ This defects are generally found in ionic crystal and are usually produce as result of thermal vibration. As tempr rises more and more schottkey defect produce in the crystal.

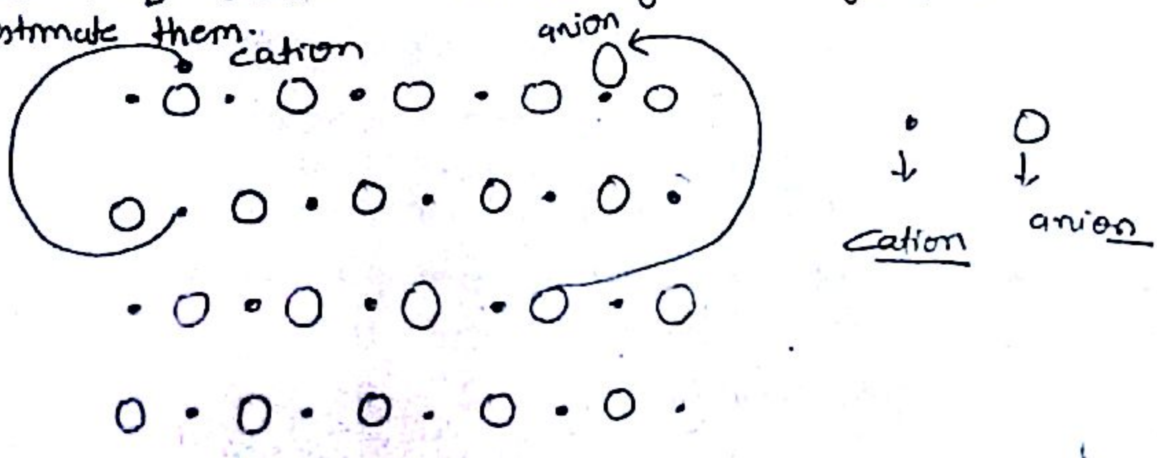


Migrated on the surface of the crystal

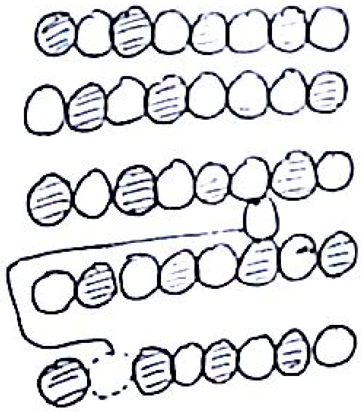
⇒ In this type of defect vacancy are created in a crystal due to the migration of ion pair to the surface of the crystal. Such defect always occurs near the surface of the crystal.

→ These defect might be produce by plucking interior ions out of its regular sites and placing it on the surface. The process requires energy. Its create disorderness that's why entropy ↑es.

→ In thermal eq^b a certain no. of vacancy are always present and we can estimate them.



⇒ **Frankel Defects** :⇒ In this type of defects, an atom or ion is transferred from its Lattice position to interstitial site, a position which is not occupied by an atom, clearly a charged neutrality is maintain by having an anion in the interstatial position. This pair of vacancy & interstaticity is called frankal defect



⇒ Concⁿ of frankal pair :⇒

Let N - ion pair

n - Frankal defects

(n anions or cations n vacancy)

If W is the different no of ways in which frankal defects are arranged in the crystal.

$$W = \frac{N!}{(N-n)!\,n!} \times \frac{N_i\,!}{(N_i-n)!\,n!} \quad \text{——— (1)}$$

$N \rightarrow$ no. of ion pair

$N_i \rightarrow$ intestatiral position of ions

Using Boltzman's Entropy relationship

$$S = k_B \log_e W \quad \text{——— (2)}$$

$$\therefore \quad S = k_B \log_e \left[\frac{N!}{(N-n)!\,n!} \times \frac{N_i\,!}{(N_i-n)!\,n!} \right] \quad \text{——— (III)}$$

→ If E_i is the energy required to move each frankal pair defects

$$\therefore \quad E = nE_i \quad \text{——— (IV)}$$

By Gibb's free energy relationship

$$F = E - TS \quad \text{——— (V)}$$

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$$nE_i - k_B T \log_e \left[\frac{N!}{(N-n)!\,n!}\right]\left[\frac{N_i!}{(N_i-n)!\,n!}\right] \quad \text{—— (6)}$$

In thermal eqb

$$\left(\frac{\partial F}{\partial n}\right)_T = 0$$

from (6) Ac to Sterling approx

$$0 = E_i - k_B T \log_e\left[\frac{(N-n)(N_i-n)}{n^2}\right] \quad \text{—— (7)}$$

Since $N >> n$

$N_i >> n$

from (7)

$$\frac{E_i}{K_B T} = \log\left(\frac{N N_i}{n^2}\right)$$

$$\therefore \frac{E_i}{K_B T} \simeq \log_e(N N_i) - 2\log_e n$$

Rearranging

$$\therefore \log_e n \simeq \left[\frac{1}{2}\log_e(N N_i) - \frac{E_i}{2K_B T}\right]$$

$$\therefore \boxed{n \simeq (N N_i)^{\frac{1}{2}}\, e^{-E_i/2K_B T}} \quad \text{—— (8)}$$

concn of Frenkel defects

→ This eqn shows that Frenkel defects are strictly temp dependent even at room temp. Some Frenkel defects are always present.
This Eqn shows that the temp [illegible] the concn of Frenkel defects [illegible] exponentially.

... no. of schottky defect in the ionic crystal.

If N is the no. of cation anion pairs

If n is the vacancy for cation

If n is the vacancy for anion

$\therefore$ No. of different ways

$$W = \left[\frac{N!}{(N-n)!\,n!}\right]\left[\frac{N!}{(N-n)!\,n!}\right]$$

Combined no of ways to create vacancy in the ionic crystal

for cation: the probability of vacancy created by the cation

for anion: the probability of vacancy created by the anion

$$W = \left[\frac{N!}{(N-n)!\,n!}\right]^2 \quad \text{--- (1)}$$

By Boltzman's entropy relationship

$$S = k_B \log_e W \quad \text{--- (2)}$$

where k_B = Boltzman's const
$= 1.38 \times 10^{-23}\ J/°K$

S = change in Entropy

$$S = k_B \log_e \left[\frac{N!}{(N-n)!\,n!}\right]^2 \quad \text{--- (3)}$$

E_s is avg. energy req. to produce a schottky defect i.e a pair of vacancy in the Interior of the crystal

$\therefore$ Total Energy required = $n E_s$

This energy changes the internal energy of the Lattice and free energy also change

$$F = E - TS \quad \text{--- (4)}$$

where free

But $E = nE_s$ --- (5)

$$F = nE_s - T k_B \log_e \left[\frac{N!}{(N-n)!\,n!}\right]^2 \quad \text{--- (6)}$$

using stirling's approx

$$\log_e x! = x\log_e x - x.$$

[illegible]

$$= 2\left(N\log_e N - N - (N-n)\log_e(N-n) \ldots - n\log_e n\right)$$

$$= 2\left(N\log_e N - (N-n)\log(N-n) - n\log_e n\right)$$

from (4)

$$F = n\bar{E_s} - 2K_BT\left[N\log_e N - (N-n)\log_e(N-n) - n\log_e n\right] \text{ --- } (7)$$

for thermal Equilibrium

$$\left(\frac{\partial F}{\partial n}\right)_T = 0$$

$$0 = \bar{E_s} - 2K_BT\left(0 - 0 + \log_e(N-n) - \log_e n\right)$$

$$= \bar{E_s} - 2K_BT\log_e\left(\frac{N-n}{n}\right)$$

$$\bar{E_s} = 2K_BT\log_e\left(\frac{N-n}{n}\right)$$

$$\log_e\left(\frac{N-n}{n}\right) = \frac{\bar{E_s}}{2K_BT}$$

$$\therefore \boxed{\frac{N-n}{n} = e^{\frac{\bar{E_s}}{2K_BT}}} \text{ --- } (8)$$

Approx In General $N >> n$

$$\therefore N-n \simeq N$$

$$\frac{N}{n} = e^{\frac{\bar{E_s}}{2K_BT}}$$

$$\boxed{n = N e^{-\frac{\bar{E_s}}{2K_BT}}} \text{ or } \boxed{n = N\exp\left[\frac{-\bar{E_s}}{2K_BT}\right]} \text{ --- } (9)$$

VImp

This eq^n (9) gives the no. of schottkey defect in the ionic crystal at ordinary crystal.

→ At any temp with the help of this eq^n we can calculate the con^n of schottkey defect. This eq^n is an called con^n of schottkey defect

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